MISSION DATE	LIFTOFF	COORDINATES
October 10–11	8:00 AM	ESPRIT

ATEC'26

A MISSION BEYOND LIMITS

SPECIFICATION BOOK

AESS | MTT-S IEEE MICROWAVE THEORY & TECHNOLOGY SOCIETY

ESPRIT STUDENT BRANCH JOINT CHAPTER

ESPRIT IEEE Student Branch

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CHALLENGE THEME:

PROJECT C.R.A.T.E.R :

CONSTRAINED RECONNAISSANCE, AUTONOMY & TELEMETRY IN EXTREME REGIONS



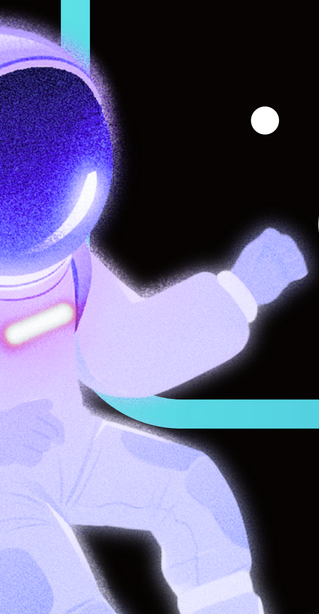
THE CHALLENGE: CONTEXT & IMPACT

As humanity returns to the Moon, missions like NASA’s CADRE (Cooperative Autonomous Distributed Robotic Exploration) are sending small autonomous rovers to map permanently shadowed lunar craters. There, standard Earth infrastructure (GPS, cellular networks, cloud computing) does not exist.

Inside a crater, lunar regolith absorbs high-frequency radio signals. If a rover finds a hazard or an ice deposit during a blackout, it cannot stream heavy data back to base.

To survive, autonomous space systems need resilient edge ai and software that can:

- Process raw visual sensor data locally
- Turn those insights into spatial coordinates
- Compress telemetry into minimal data payloads
- Relay it through lossy networks back to Mission Control



2. THE MISSION:

Propose and prototype a software system that lets autonomous lunar scouts detect environmental hazards, estimate their locations, transmit telemetry over constrained networks, and show actionable data to Mission Control.
The Four Core Pillars

- **Perceive: Process visual sensor feeds locally to detect surface hazards in real time.**
- **Localize: Convert visual detections into grid coordinates relative to the landing site.**
- **Transmit: Compress and send telemetry over an emulated low-bandwidth, lossy network.**
- **Visualize: Build a Mission Control dashboard that shows incoming spatial data and how fresh it is.**

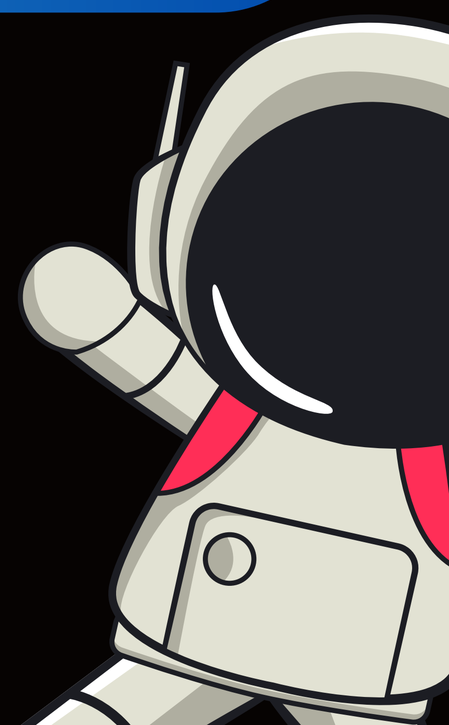


Fixed :

- 100% software-emulated format
- Core data flow: Perception → Telemetry → Dashboard
- Focus on operating under severe bandwidth constraints

Flexible:

- AI models
- Serialization methods
- Network protocols
- Mathematical approaches
- Visualization style
- Programming languages



3. CORE FUNCTIONAL OBJECTIVES:

Teams must demonstrate a defensible systems architecture through a fully team-implemented software pipeline.

- **Onboard Perception: Detect key features or hazards from visual feeds locally, with no external cloud APIs.**
- **Spatial Awareness: Project detections into spatial coordinates relative to a reference origin.**
- **Low-Bandwidth Communication: Serialize and send coordinate telemetry in ultra-compact formats suited for degraded radio links.**
- **Resilient Data Transport: Successfully route and deliver telemetry through team-implemented network sockets or scripts simulating signal dropouts and packet loss.**
- **Mission Control Interface: Show incoming spatial insights on a clear dashboard that indicates data age or uncertainty to operators.**

4. THE ARCHITECTURE TOOLBOX:

Teams are free to choose their technical stack, as long as every component is justified by the mission constraints:

- **Perception: Object detection models, classical computer vision, or lightweight custom classifiers.**
- **Spatial Math: Kinematic transforms, geometry projection, or frame-transformation libraries (e.g., tf2).**
- **Serialization & Data Flow: Compact binary structs, custom byte arrays, or lightweight protocols.**
- **Dashboard & UI: Web frameworks (React, Vue, HTML/JS), visualization libraries, or custom operator interfaces.**

5. EVALUATION PROCESS & DELIVERABLES

The competition is structured into two distinct evaluation stages:

Phase 1: Technical Preselection

All registered teams will submit their code repository and automated test outputs at the code freeze deadline. The technical jury will benchmark and grade each pipeline across the five core technical categories.
The top 7 highest-scoring teams from this preselection phase will be selected as Finalists to advance to the live pitching stage.

Phase 2: Live Pitching & Q&A Stage

The 7 Finalist teams will deliver a live 7-minute technical presentation to the grand jury, followed by a 3-minute technical Q&A session. They must demonstrate their end-to-end pipeline running live and defend their architectural and mathematical choices.



JUDGING RUBRIC

Stage	Category	**Weight**	Evaluation Focus
Preselection	Edge Perception Efficiency	15%	Model speed (FPS), tracking accuracy, and resource efficiency on constrained compute.
Preselection	Spatial & Kinematic Math	15%	Accuracy and logical defense of the 2D pixel to 3D spatial coordinate transform.
Preselection	Telemetry & Protocol Efficiency	15%	Creativity in byte compression/ serialization and payload survival through lossy links.
Preselection	Mission Control Dashboard	15%	Real-time data ingestion, UI clarity, and representation of data aging/ uncertainty.
Preselection	Systems Integration & Space Compliance	15%	Seamless automated data flow from video ingestion to dashboard output and adherence to space flight-software principles .
Finals	Pitching & Technical Q&A	25%	Clarity of pitch presentation, live demonstration quality, and ability to defend architectural choices during Q&A.

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THANK YOU FOR YOUR PARTICIPATION

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